Supplementary Material for the paper:

Binaural Acoustics Sensing for Speech Intelligibility Assessment: A Dual-Branch CNN–MLP Framework Using Binaural Room Impulse Responses

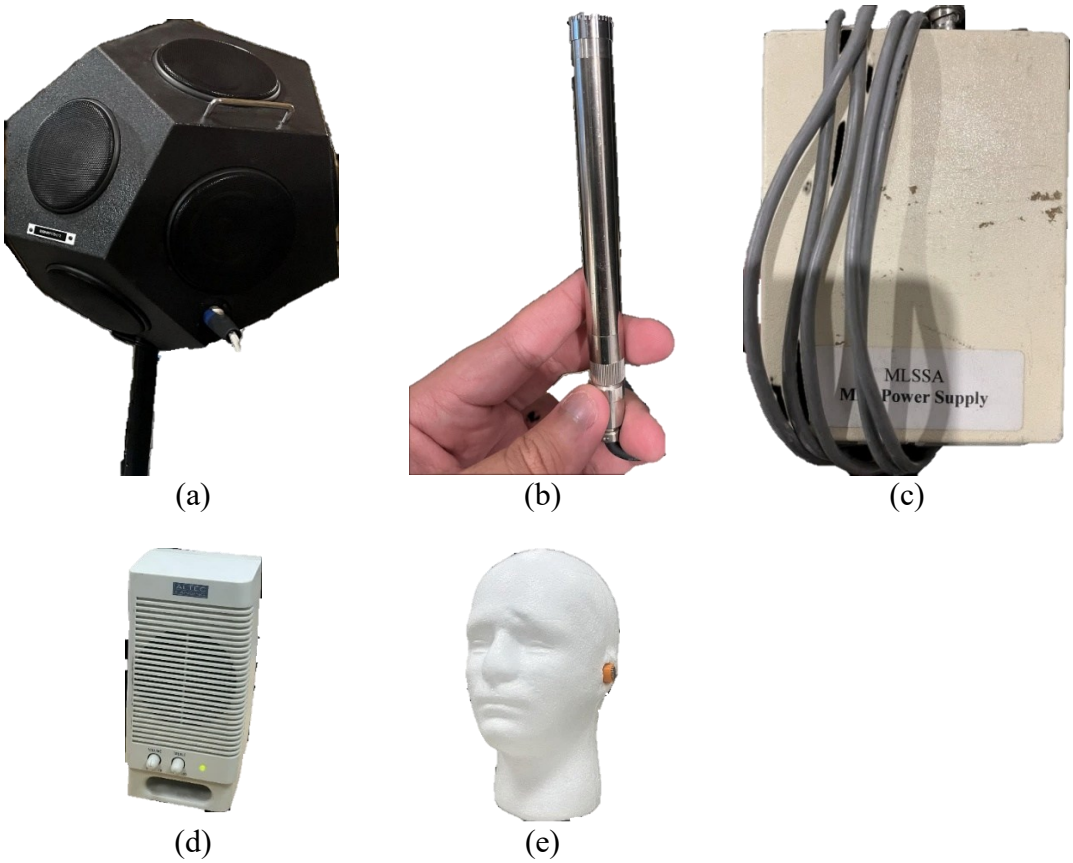
(a)

(b)

Supplementary Figure 1. *Field measurement location (Al-Wable Mosque, Saudi Arabia): (a) Interior view showing a large central dome, four columns as support, an open prayer hall layout, and decorative ceiling lighting, (b) Exterior view of the mosque showing external stone facade and dome illumination enhances its visibility and aesthetic prominence during nighttime (source: Google).*

***Supplementary Table 1**. Architectural characteristics of Al-Wabel Mosque.*

Name	Al-Wabel Mosque
Location	Dammam, Saudi Arabia
Function	Daily and Friday prayers
Length (L) x Width (W) x Height	36 x 36 x 7 m
Aspect ratio (L/W)	1
Floor area	1260 m^2
Number of columns and domes	4 and 1, respectively
Horizontal and Vertical diameters of the dome	14 m and 7 m, respectively



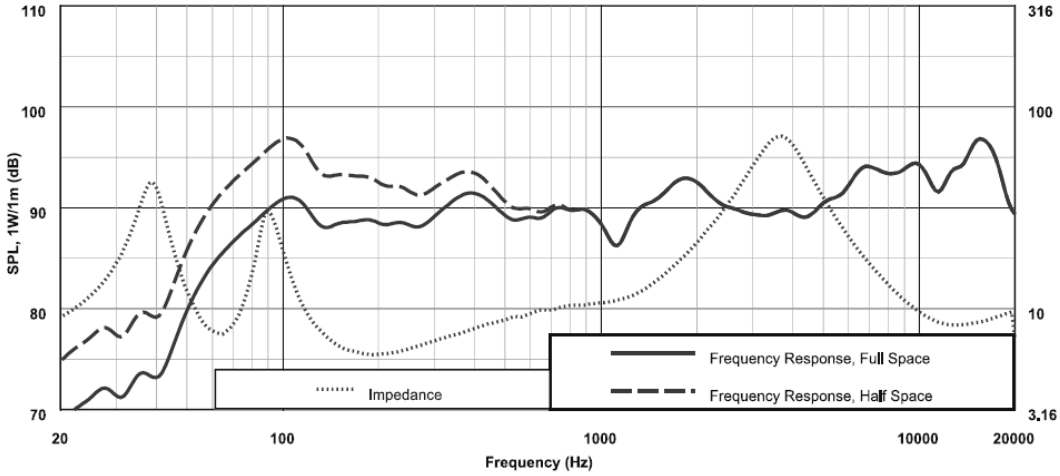
(a) (b) (c)

(d) (e)

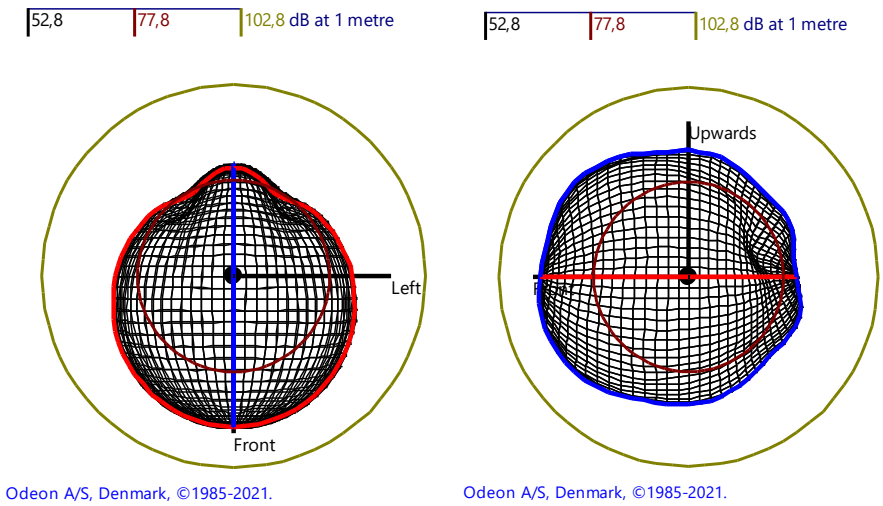
***Supplementary Figure 2**. Equipments used during field measurement: (a) An omnidirectional source from 01dB that consists of a 12 loudspeaker-dodecahedron and contains a power amplifier and a noise generator. (b) A ½ inch condenser microphone from Brüel & Kjær that has a frequency response range of 5 Hz to 20kHz [168]. (c) An external microphone's amplifier. (d) A unidirectional loudspeaker from Altec Lansing (ACS 90 type) that mimics Imam's voice during preaching. (e) A*

dummy head made of foam with holes (lined with PLA material) designed to hold microphones in the ear position.

(a)

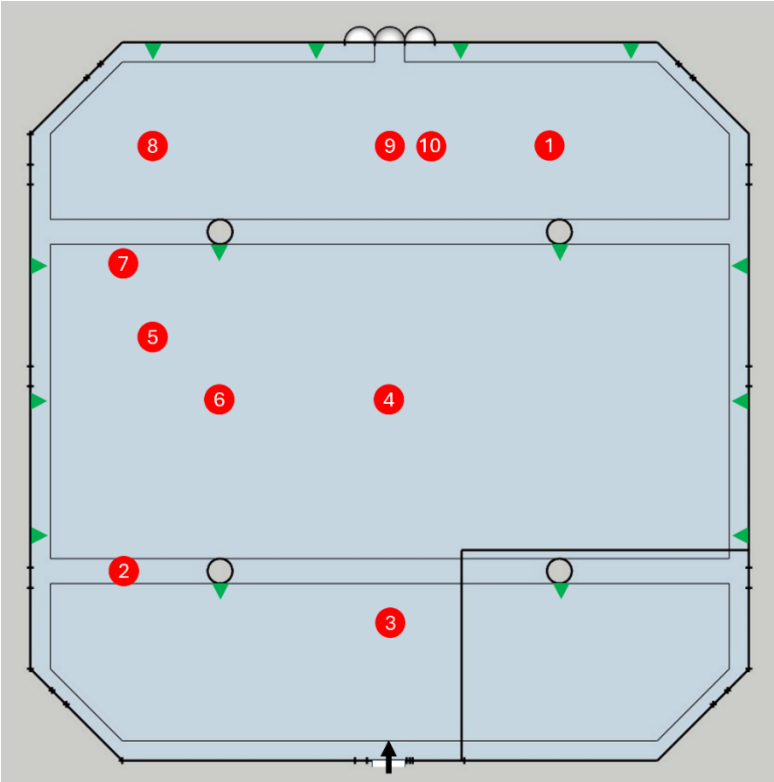


(b)



(c)

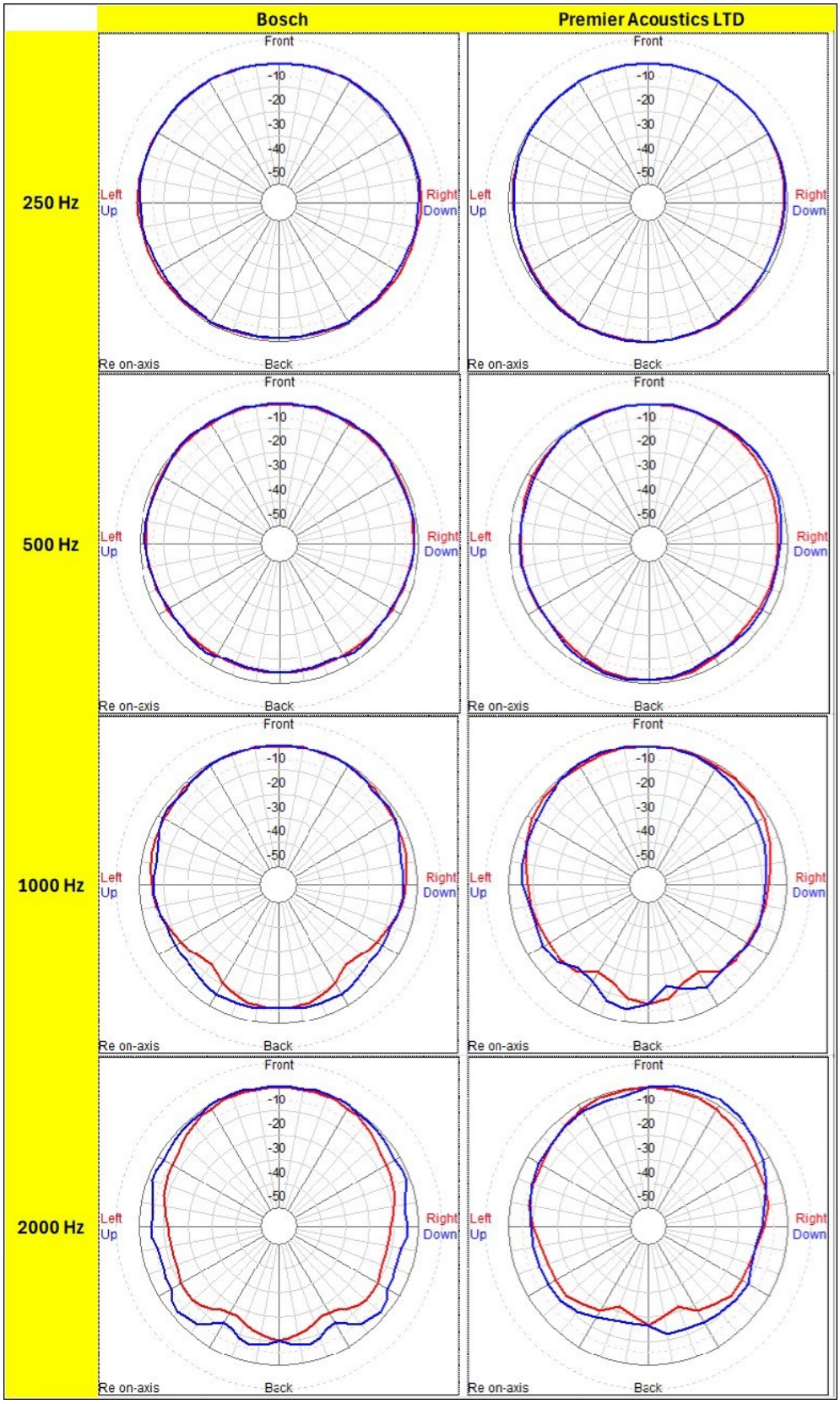
Supplementary Figure 3. *Technical specifications of the existing mosque's loudspeaker, ElectroVoice Zx1. (a) Physical form of the 8-inch loudspeaker unit. (b) Frequency response and impedance curves of the loudspeaker [171]. (c) Directivities of the loudspeaker: the left diagram shows the horizontal dispersion, while the right diagram shows the vertical dispersion.*



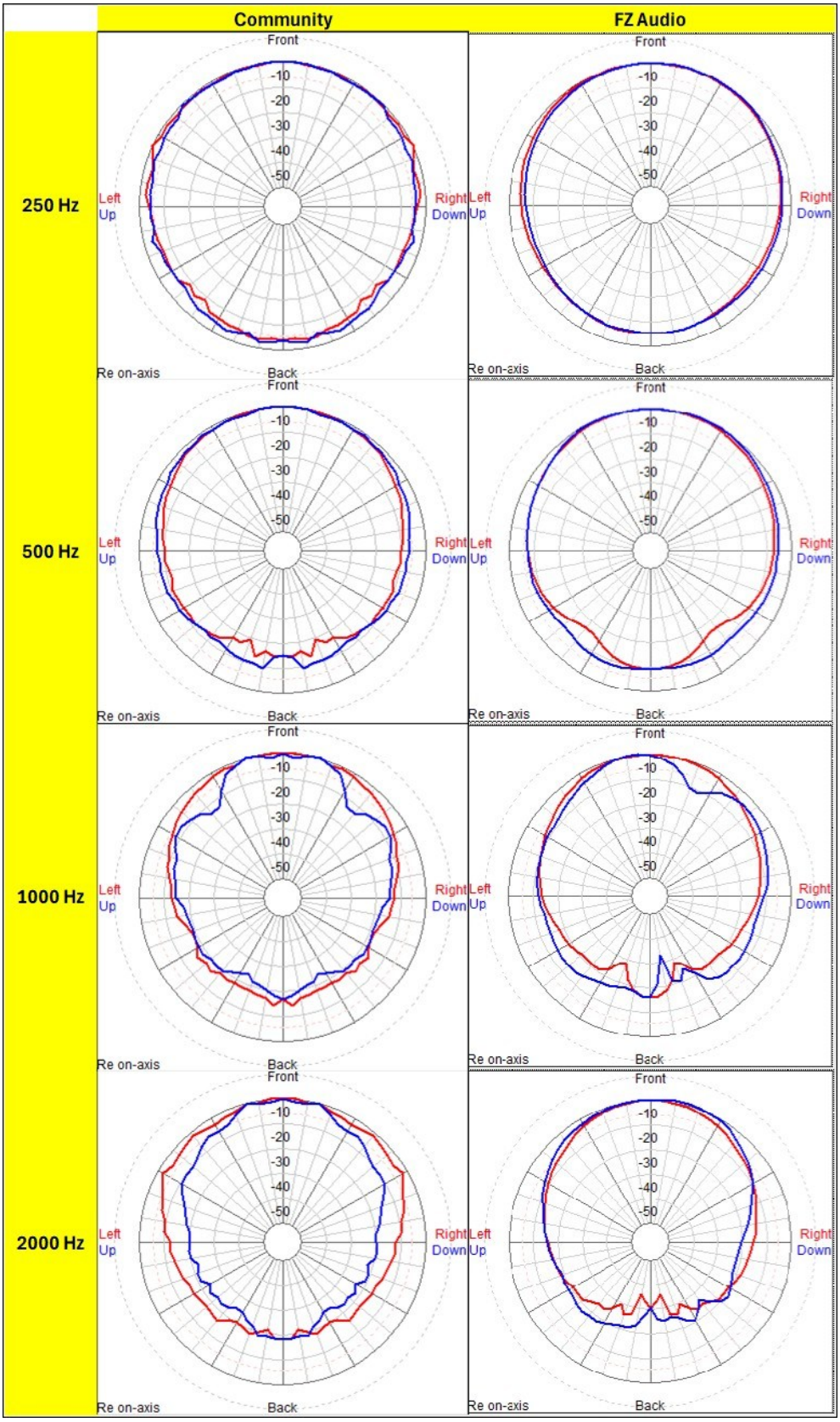
Supplementary Figure 4. *Field measurement setup showing 10 measurement positions. Symbol key: Red circle = measurement position; green triangle = installed mosque loudspeaker; and black arrow = entrance. The qibla direction is toward the qibla niche.*

Supplementary Figure 5. *Set up of binaural acoustic measurements using a dummy head inside the mosque. A researcher positions a 1/2-inch condenser microphone into the dummy head's ear canal,*

which is mounted on a tripod at listener ear height (0.85 m) to simulate a seated listener during khutbah delivery.

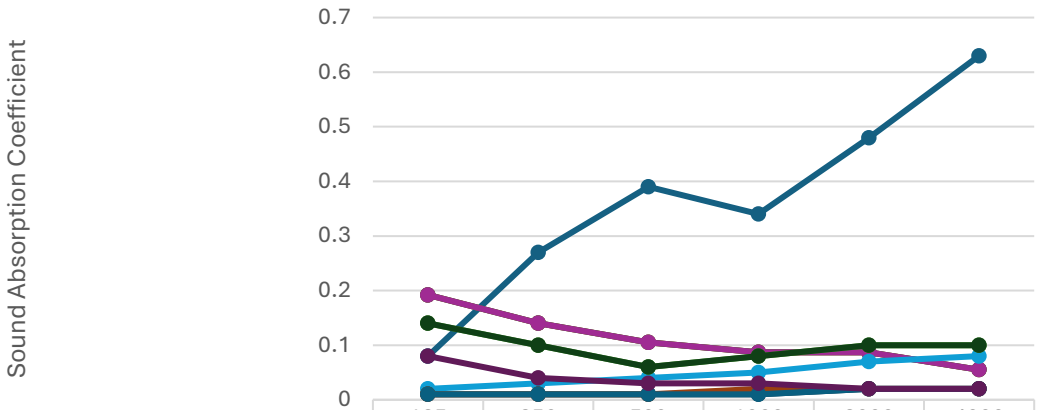


(a)



(b)

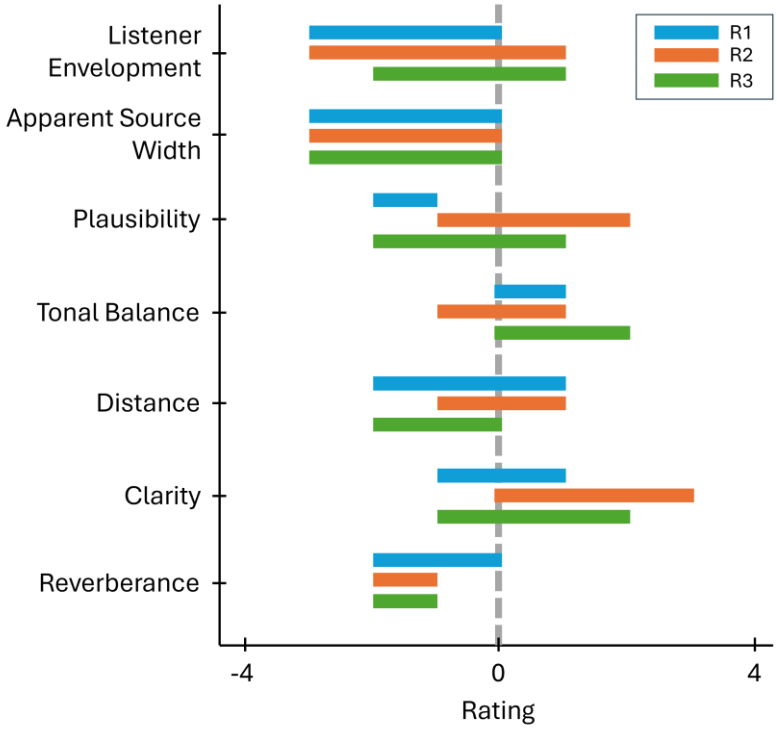
Supplementary Figure 6. *Directivities of loudspeakers at octave band mid frequencies of 250 Hz, 500 Hz, 1000 Hz, and 2000 Hz from four manufacturers: (a) Bosch and Premier Acoustics LTD; (b) Community and FZ Audio. Each plot shows the directivity with red lines indicating horizontal directivity and blue lines indicating vertical directivity.*



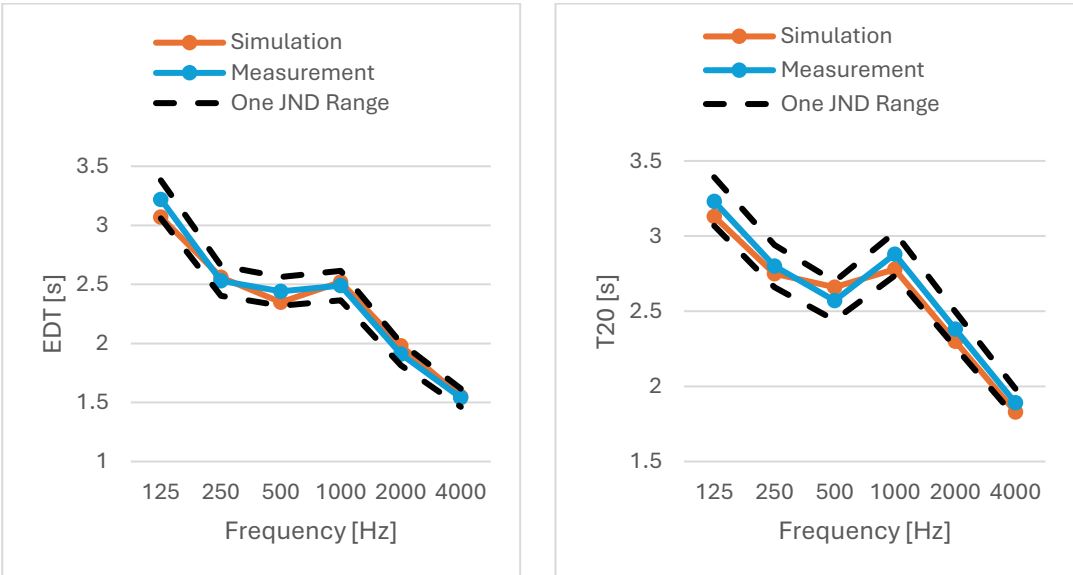
	125	250	500	1000	2000	4000
Carpet - Carpet heavy, on hairfelt or foam rubber	0.08	0.27	0.39	0.34	0.48	0.63
Ceiling - Plaster decorative	0.192	0.14	0.105	0.087	0.087	0.055
Column - Plaster decorative	0.192	0.14	0.105	0.087	0.087	0.055
Wall - Plaster decorative	0.192	0.14	0.105	0.087	0.087	0.055
Dome - Plaster with wallpaper	0.02	0.03	0.04	0.05	0.07	0.08
Door - Solid wooden	0.14	0.1	0.06	0.08	0.1	0.1
Floor tile - Marble or glazed tile	0.01	0.01	0.01	0.01	0.02	0.02
Mihrab - Smooth concrete, painted	0.01	0.01	0.01	0.02	0.02	0.02
Partition - Solid wooden	0.14	0.1	0.06	0.08	0.1	0.1
Wallbase - Marble or glazed tile	0.01	0.01	0.01	0.01	0.02	0.02
Windows - Single pane of glass, 3 mm	0.08	0.04	0.03	0.03	0.02	0.02

Frequency [Hz]

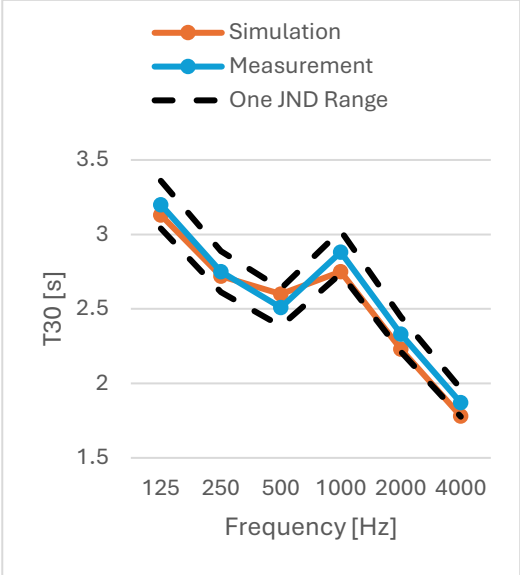
***Supplementary Figure** 7. Sound absorption coefficients of materials applied to the surfaces. Floor tile refers to the floor area not used for praying. Wallbase refers to the wall area located 1 meter above the floor.*



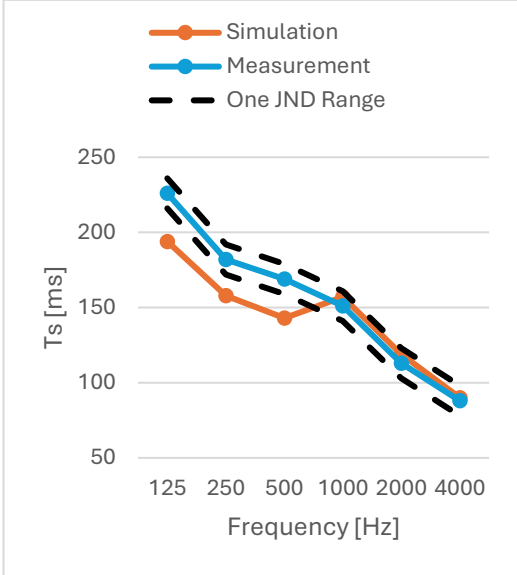
Supplementary Figure 8. *Subjective comparison of measured and simulated mosque audio recordings. The confidence intervals for seven acoustic metrics are shown for three positions: R1 (near the rear wall), R2 (near the side wall), and R3 (near the front wall). Positive ratings favor the measured recording, while negative ratings favor the simulated recording. The result is considered neutral (no differences between the two recordings) when the confidence intervals include 0.*



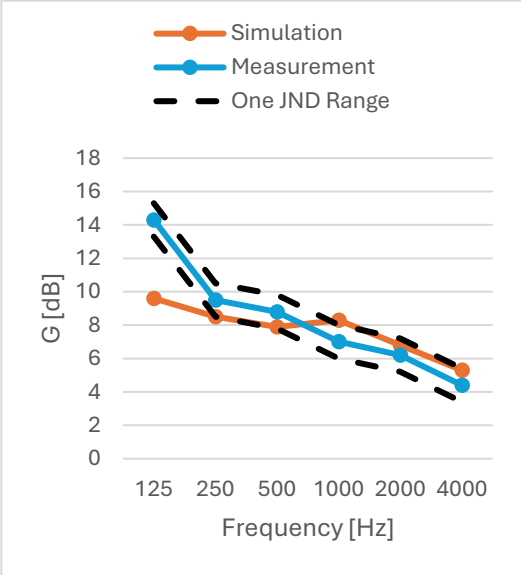
(a) (b)


Simulation
Measurement
One JND Range
T30 [s]
3.5
3
2.5
2
1.5
125 250 500 1000 2000 4000
Frequency [Hz]

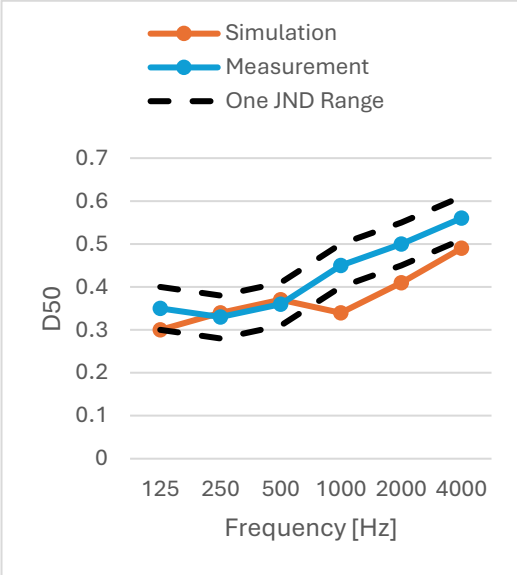
(c)


Simulation
Measurement
One JND Range
Ts [ms]
250
200
150
100
50
125 250 500 1000 2000 4000
Frequency [Hz]

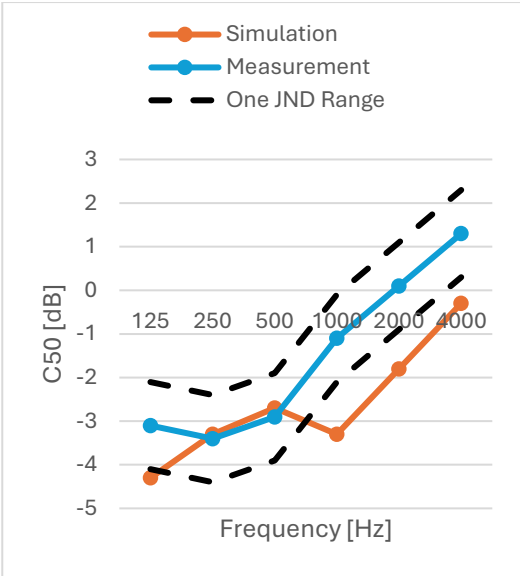
(d)


Simulation
Measurement
One JND Range
G [dB]
18
16
14
12
10
8
6
4
2
0
125 250 500 1000 2000 4000
Frequency [Hz]

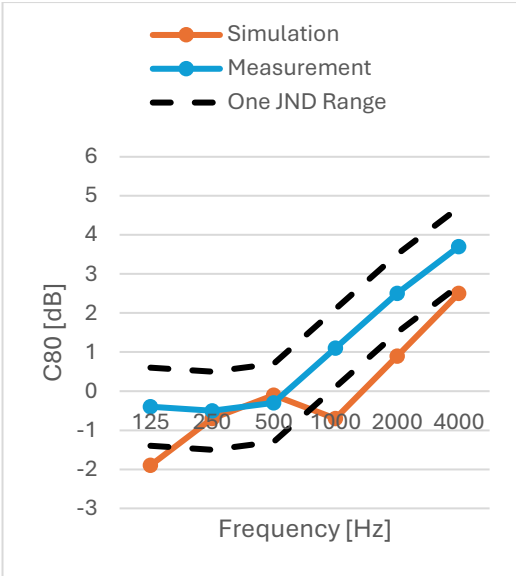
(e)


Simulation
Measurement
One JND Range
D50
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
125 250 500 1000 2000 4000
Frequency [Hz]

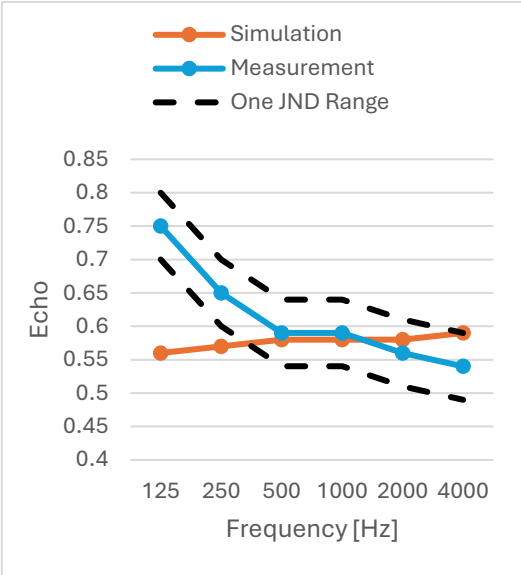
(f)



(g)



(h)



(i)

Supplementary Figure 9. *Comparison result of nine objective metrics between measured and simulated RIR for calibration: (a) EDT, (b) T20, (c) T30, (d) Ts, (e) G, (f) D50, (g) C50, (h) C80, and (i) Echo. Most of the metrics are within the one JND range, indicating that the simulation model is considered successfully calibrated.*